

# SUPPORT2 Dataset: Data Cleaning & Preparation

AAI-500 Applied Statistics for AI | Final Team Project

---

## Notebook Objectives

This notebook covers the **Data Cleaning & Preparation** phase of the project pipeline:

1. Initial data inspection and profiling
  2. Column renames per PEP-8
  3. Engineering the binary target variable ( `death_180d` )
  4. Data type casting and ordinal/nominal encoding
  5. Missing data analysis and imputation
  6. Dropping unused columns
  7. Documenting target / feature / outcome
  8. Export of the cleaned dataset for EDA and modeling
- 

## Section 0: Setup & Imports

---

## Section 1: Load Data & Initial Inspection

### 1.1 Descriptive Statistics

---

## Section 2: Column Renaming

Rename some columns to follow PEP-8 standard

---

## Section 3: Target Variable

## Did the patient die within 180 days of study?

To answer this we look at two fields:

- `death` : whether the patient ever died during 180 days (1 = yes, 0 = no)
- `d_time` : days of follow-up (how long they were observed). If this number is greater than 180 days of the trial, that means they survived the trial

`death_180d` (1 = yes, 0 = no) = {`death=1 AND d_time <= 180`}

Notice the AND conditional because the patient must've died within the 180 days to satisfy our target. We then use **`death_180d`** for binary classification

---

## Section 4: Data Type Casting (into Pandas Type)

Correct data types are essential for proper computing and modeling. We apply three categories of correction. Casting to `Int8` also handles NaN gracefully:

1. **Binary integer columns** -> `Int8` (nullable integer; memory-efficient)
2. **Nominal categorical columns** -> unordered `pd.Categorical`
3. **Ordinal categorical columns** -> ordered `pd.Categorical` for proper sorting and comparison

**income ordering (ascending socioeconomic status):** `under $11k < $11-$25k < $25-$50k < >$50k`

**sfdm2 ordering (best functional status → worst):** `no disability < moderate ADL impairment < severe SIP impairment < coma/intubated < died before 2-month interview`

---

## Section 5: Missing Data Analysis

The authors have stated there are a maximum of 5641 N/As. This, along with every other missing data, we must verify and address.

### 5.1 Missing Data Decision Table

Here we decide how to tackle missing data. Either drop the column entirely or impute mathematically

For most of our biological features that have missing data, we compute the median. The reason is per Agresti et al (2022), chapter 3, that for right-skewed data, using median instead of mean worse the skewness of the data (push it toward the tail). Looking at section 1, our columns the mean > median, which signals right skewed.

**Column(s)    Decision    Rationale**

| `adlp` , `adls` | **Drop** | Superseded by `adlsc` per original authors | `dnrday` | **Drop** | Do-no-resus day. Useless for our purpose | `hday` | **Drop** | Day admitted to study. Nothing to do with mortality | `charges` , `totcst` , `totmcst` | **Drop** | Relating to hospital cost, which we won't be studying | `alb` , `pafi` , `bili` , `crea` , `bun` , `wblc` , `urine` | **imputed values avail** | Per Author's note we have the recommended imputed values | `ph` , `glucose` , `scoma` , `avtisst` , `edu` , `prg2m` , `prg6m` | **Sample median** | high missing | | `surv2m` , `surv6m` , `aps` , | **Sample median** | kept as benchmark columns | | `meanbp` , `hrt` , `resp` , `temp` , `sod` | **Sample median** | few missing | | `charges` , `totcst` | **Sample median** | Retained for future cost analysis | | `totmcst` | **Sample median** | Retained for future cost analysis; many missing values and will be difficult to impute. Contains negative values (balances) | | `race` , `income` , `sfdm2` , `dnr` | **Mode imputation** | Most frequent category as best single-value estimate for categoricals |

Benchmark Columns (kept but will be excluded from death\_180d model features)

| Column                                    | Role                         | Notes                                                                                                     |
|-------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------|
| <code>surv2m</code> , <code>surv6m</code> | SUPPORT model predictions    | will be compared against our model                                                                        |
| <code>aps</code>                          | APACHE III severity score    | Independent benchmark system widely used in critical care                                                 |
| <code>sps</code>                          | SUPPORT physiology sub-score | Component of the SUPPORT model ??                                                                         |
| <code>prg2m</code> , <code>prg6m</code>   | Physician survival estimates | ML vs. clinician judgment comparison                                                                      |
| <code>dnr</code>                          | DNR order status             | <b>Should not use as predictor feature as it's reflects human behavior rather than biological factors</b> |

## Section 6: Feature Pruning

We remove features not used in our analysis:

## Section 7: Missing Data Imputation

The original SUPPORT study authors (Harrell, 2022) provide clinically validated **normal fill-in values** for seven physiological variables (see docs/support2\_dataset\_description.md). We will use this for imputation

| Variable | Normal Fill | Meaning                           |
|----------|-------------|-----------------------------------|
| alb      | 3.5         | Serum albumin - liver / nutrition |
| pafi     | 333.3       | PaO2/FiO2 ratio - oxygen level    |
| bili     | 1.01        | Bilirubin - liver function        |
| crea     | 1.01        | Creatinine - renal function       |
| bun      | 6.51        | BUN - normal renal function       |
| wbhc     | 9.0         | WBC count - immune status         |
| urine    | 2502        | Urine output - renal output       |

---

## Section 8: Defining Feature, Outcome Separation, Leakage

| Category                                         | Columns                          |
|--------------------------------------------------|----------------------------------|
| Target                                           | death_180d                       |
| Outcome columns (excluded due to result leakage) | death , hospdead , d_time , slos |
| Row identifier                                   | id                               |
| Features                                         | All remaining columns            |

---

## Section 9: Post-Cleaning Summary

We compare the dataset before and after cleaning and present the five-number summary for key physiological lab variables to confirm that imputation and outlier retention produced a coherent, analysis-ready dataset.

### 9.1 Important Lab Variables Summary (Post-Cleaning)

---

## Section 10: Save Cleaned Dataset

# SUPPORT2 Dataset: Exploratory Data Analysis

**AAI-500 Applied Statistics for AI | Final Team Project**

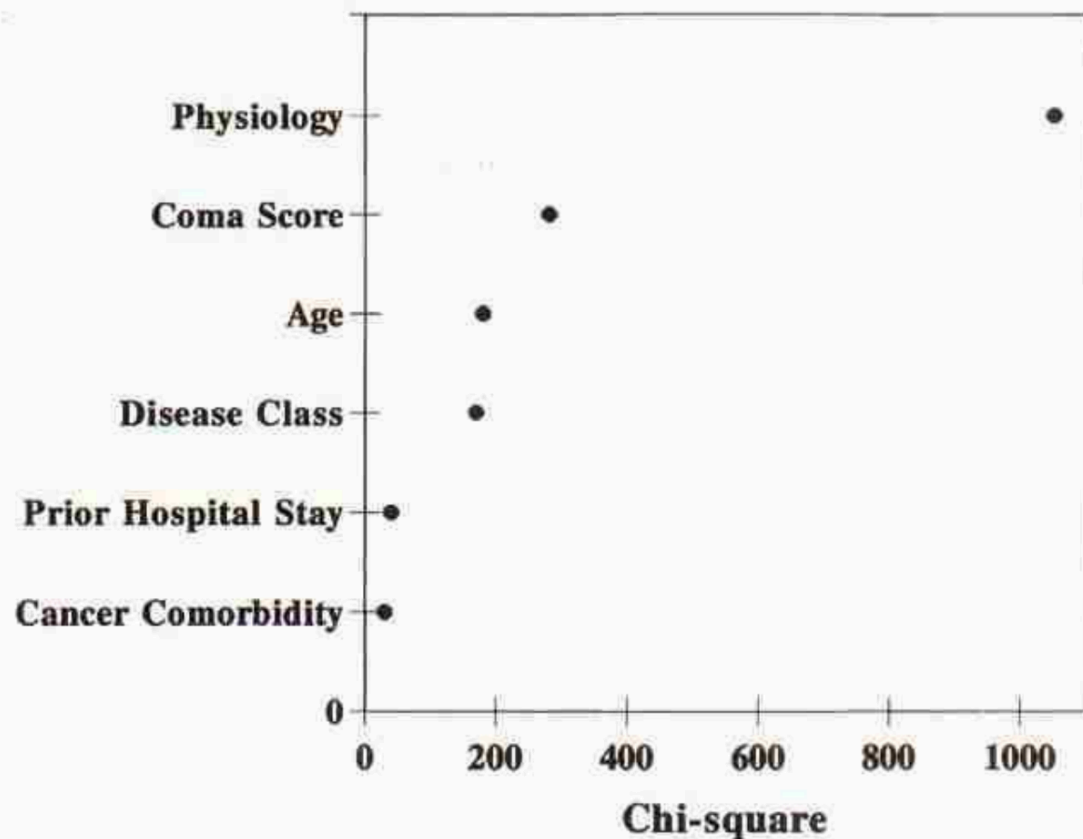
---

## Notebook Objectives

This notebook covers the **Exploratory Data Analysis** phase of the project pipeline:

1. Detect Outliers
  2. Univariate Analysis
  3. Bivariate Analysis
  4. Demographic Analysis
  5. Disease Group Analysis
  6. DNR Status Analysis
- 

## Learnings from the 1990s study



**Figure 1. Relative contributions of major prognostic elements in the SUPPORT model as measured by the amount of chi-square accounted for by each element.** Physiology = all physiologic measures except Glasgow coma scale; cancer comorbidity = presence of cancer in addition to disease category; previous hospital stay = days in the hospital before study entry.

To inform our work, we review Knaus et al's (1995) paper. According to Knaus et al these features were most weighted in their model:

- Physiology Score (calculated from wbc, albumin, etc). Some salient ones are Blood Pressure and Oxygenation levels
- Neurological Function (Glasgow coma scale)
- Patient age (but only when coupled with certain diseases like COPD)
- Apache III score
- Disease-Specific Interactions: The mathematical weight of certain features was explicitly programmed to vary based on the patient's primary disease: Albumin x Disease: A low albumin level strongly predicted death in cancer, COPD, and congestive heart failure, but was relatively unimportant in coma or MOSF. WBC x Disease: A low WBC count was specifically associated with a greater risk of death in acute respiratory failure and MOSF. Age x Disease: Advancing age heavily impacted short-term survival in

COPD but had almost no incremental effect on patients with severe acute complications like MOSF or malignancy.

---

## Section 0: Setup & Imports

---

## Section 1: Load Cleaned Data

---

## Section 2: Physiology Outlier Detection (IQR Method)

$IQR = Q3 - Q1$

$Lo\ whisker = Q1 - 1.5IQR$

$Hi\ whisker = Q3 + 1.5IQR$

- Urine boxplot median is pushed all the way right because of imputation with Norm-Fill.
  - Glucose boxplot collapses due to high number of imputed NAs.
  - **Probably better to run outliers before imputation**
- 

## Section 3: Univariate Distributions

- Lab features are mainly right skewed
- Some imputations cause spikeness (glucose, pafi, bili)

### 3.1 Age follows normal distribution

---

## Section 4: Bivariate Analysis - Lab Values by Survival Outcome

- Difficult to see any impact these lab values make with this analysis
- The paper did say this only becomes evident when combined with Disease group. For example, a high creatine with kidney failure gives high likelihood of mortality.

---

## Section 5: Demographic Analysis

- Mortality gradual decline as wealth increases
- Asian group a bit more mortality

```
# education bins
edu_bins = [0, 8, 12, 15, 100]
edu_labels = ["0-8 yrs", "9-12 yrs", "13-15 yrs", "16+ yrs"]
df["edu_group"] = pd.cut(df["edu"], bins=edu_bins, labels=edu_labels,
right=True)

fig, axes = plt.subplots(2, 3, figsize=(18, 10))

plot_mortality_rate(
    df, "edu_group", TARGET_COL, overall_rate, axes[0, 0],
    title="Education", sort=False
)
plot_mortality_rate(df, "income", TARGET_COL, overall_rate, axes[0, 1],
    title="Income")
plot_mortality_rate(df, "race", TARGET_COL, overall_rate, axes[0, 2],
    title="Race")
plot_mortality_rate(df, "sex", TARGET_COL, overall_rate, axes[1, 0],
    title="Sex")
plot_mortality_rate(
    df, "ca", TARGET_COL, overall_rate, axes[1, 1], title="Cancer Status"
)
axes[1, 2].set_visible(False)

fig.suptitle(
    "Section 5: 180-Day Mortality Rate by Demographic & Categorical
Features",
    fontsize=12,
    fontweight="bold",
)
plt.tight_layout()
plt.show()

# Survivorship by age
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

sns.boxplot(data=df, x=TARGET_COL, y="age", ax=axes[0], palette=["green",
"red"])
axes[0].set_title("Age by Survival Outcome")
axes[0].set_xticklabels(["Survived", "Died"])
axes[0].set_xlabel("")
axes[0].set_ylabel("Age (years)")

# KDE overlay
```



```

for label, grp, color in [("Survived", survivors, "green"), ("Died", died,
"red")]:
    grp["age"].plot.kde(ax=axes[1], label=label, color=color, linewidth=2)
axes[1].axvline(survivors["age"].mean(), color="green", linestyle="--",
linewidth=1)
axes[1].axvline(died["age"].mean(), color="red", linestyle="--",
linewidth=1)
axes[1].set_title("Age Distribution by Survival Outcome")
axes[1].set_xlabel("Age (years)")
axes[1].set_ylabel("Density")
axes[1].legend()

plt.tight_layout()
plt.show()

```

---

## Section 6: Disease Group Analysis

---

## Section 7: DNR Status

- **no dnr**: no order placed
  - **dnr before sadm**: order existed before study admission
  - **dnr after sadm**: order placed during the admission (possibly due to deteriorating conditions)
- 

## Section 8: Categorical Features: Chi-Squared Tests & Contingency Tables

Tests whether a categorical variable is **statistically independent** of 180-day mortality.

$H_0$  : the variable is independent of death\\_180d

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Reject  $H_0$  when  $p < 0.05$ .

**Standardized residuals** =  $(O - E)/\sqrt{E}$  — cells with  $|r| > 2$  deviate significantly from independence and drive the chi-squared result.

## 8.1 Race and Sex shows low P significance

## 8.2 Within Disease Group and Cancer, which class affects mortality most?

- Disease Group: MOSF w/ malignant, Coma, Lung Cancer (in that order) die more than overall rate
- Cancer: metastatic patients die more than the overall rate

---

## Section 9: Additional Feature Pruning after EDA

Base on our EDA we are opting to drop the following features:

| Column(s) | Decision    | Rationale                                                  |
|-----------|-------------|------------------------------------------------------------|
| sex       | <b>Drop</b> | Weak signifance base on Bivariate and Chi-Squared analysis |
| edu       | <b>Drop</b> | Weak signifance base on Bivariate and Chi-Squared analysis |